EliteDesign: Elite-Guided Learning for RNA Design via the Cross-Entropy Method

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Abstract

The aim of RNA Design is to find, within an exponentially large search space, a nucleotide sequence that folds into a given target structure. Efficient RNA design could aid in the development of RNA-based technologies such as riboswitches and ribozymes. However, finding sequences that fold into the target structure with a high probability remains challenging, particularly for long RNAs. To address this challenge, we introduce an approach that uses the Cross-Entropy method to iteratively refine a distribution over valid RNA sequences. Our method outperforms other state-of-the-art methods, achieving an average folding probability of over 0.60 on Eterna100, with a recorded runtime of seven minutes per seed for the full benchmark. It also scales to longer RNAs, reaching an average target-structure probability of 0.51 on ten long 16S rRNA targets, compared with 0.20 for SAMFEO, while recording a sevenfold lower average runtime per puzzle.